

Linear Recursion Practice Set (20 Questions)

Integer Problems

1. Print Numbers from 1 to N

Write a recursive function to print all numbers from 1 to N in increasing order.

Sample Test Cases

Input:5 → Output:1 2 3 4 5

Input:1 → Output:1

Input:7 → Output:1 2 3 4 5 6 7

2. Print Numbers from N to 1

Write a recursive function to print all numbers from N to 1 in decreasing order.

Sample Test Cases

Input:5 → Output:5 4 3 2 1

Input:1 → Output:1

Input:3 → Output:3 2 1

3. Factorial of a Number

Write a recursive function to find the factorial of a given number.

Sample Test Cases

Input:5 → 120

Input:0 → 1

Input:6 → 720

4. Sum of First N Natural Numbers

Write a recursive function to find the sum of the first N natural numbers.

Sample Test Cases

Input:5 → 15

Input:10 → 55

Input:1 → 1

5. Sum of Digits

Write a recursive function to find the sum of all digits in a number.

Sample Test Cases

Input:5234 → 14

Input:999 → 27

Input:1005 → 6

6. Count Digits

Write a recursive function to count the digits in a number.

Sample Test Cases

Input:987654 → 6

Input:5 → 1

Input:1000 → 4

7. Reverse a Number

Write a recursive function to reverse a given number.

Sample Test Cases

Input:12345 → 54321

Input:900 → 9

Input:808 → 808

Array Problems

8. Print Array Elements

Write a recursive function to print all elements of an array.

Sample Test Cases

[5,2,8,1] → 5 2 8 1

[9] → 9

[1,2] → 1 2

9. Print Array in Reverse

Write a recursive function to print the array in reverse order.

Sample Test Cases

[5,2,8,1] → 1 8 2 5

[9] → 9

[1,2] → 2 1

10. Sum of Array Elements

Write a recursive function to calculate the sum of all elements.

Sample Test Cases

[3,5,2,4] → 14

[1] → 1

[2,2,2] → 6

11. Find Maximum Element

Write a recursive function to find the largest element.

Sample Test Cases

[3,8,1,6] → 8

[-1,-5] → -1

[7] → 7

12. Find Minimum Element

Write a recursive function to find the smallest element.

Sample Test Cases

[3,8,1,6] → 1

[-1,-5] → -5

[7] → 7

13. Count Occurrences

Write a recursive function to count occurrences of a given element.

Sample Test Cases

[2,5,2,7,2],2 → 3

[1,1,1],1 → 3

[4,5],3 → 0

14. Check Sorted Array

Write a recursive function to determine whether an array is sorted in ascending order.

Sample Test Cases

[1,2,3] → Sorted

[3,2,1] → Not Sorted

[5] → Sorted

String Problems

15. Print Characters

Write a recursive function to print each character of a string on a new line.

Sample Test Cases

Hello

A

Java

16. Count Vowels

Write a recursive function to count the number of vowels in a string.

Sample Test Cases

Programming → 3

Education → 5

Sky → 0

17. Count Character Occurrences

Write a recursive function to count occurrences of a character in a string.

Sample Test Cases

banana,a → 3

hello,l → 2

java,z → 0

18. Reverse a String

Write a recursive function to reverse a string.

Sample Test Cases

Hello → olleH

Java → avaJ

A → A

19. Check Palindrome

Write a recursive function to determine whether a string is a palindrome.

Sample Test Cases

madam → Palindrome

level → Palindrome

hello → Not Palindrome

20. Count Uppercase Letters

Write a recursive function to count uppercase letters.

Sample Test Cases

HeLLo → 3

JAVA → 4

hello → 0